

Alli-Trade

SW Engineering CSC648-848 Spring 2026

Team 17

Milestone 2

Name	Role
Subreet Singh ssingh61@sfsu.edu	Team Lead
Bikendra Shrestha	Back-End Lead
Michal Krupa	Github Maintainer
Jonathan Mai	Front-End Lead
Daniel Smirnoff	
Lakshyaraj Bhati	
Nicholas Blackson	

History Table

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1. Executive Summary

San Francisco State University students often need affordable ways to buy and sell everyday items such as textbooks, furniture, electronics, and services. While platforms like Facebook Marketplace or Craigslist exist, they are not tailored to the SFSU community and often involve risks such as scams, unreliable sellers, or meeting strangers outside the campus network. Students also frequently need services such as tutoring, moving help, or small freelance work that are difficult to advertise safely within the university community. Alli Trade is designed to address this gap by creating a dedicated marketplace exclusively for SFSU students, providing a safer, more convenient way to exchange goods and services within a trusted campus network.

Alli Trade is a web application that allows SFSU students to post items for sale, browse listings, offer services, and communicate directly with other students. The platform functions similarly to a simplified campus version of Facebook Marketplace but is built specifically for the needs of university students. Users can create listings for items they wish to sell, place bids on listings, offer services such as tutoring or design work, and communicate through an integrated messaging system. Importantly, the platform does not process payments directly; instead, it facilitates communication between buyers and sellers so they can arrange transactions independently. This approach keeps the platform simple while giving students a convenient way to connect with others on campus.

What makes Alli Trade unique is its focus on the SFSU student community. Access is limited to SFSU students, helping create a trusted environment where users know they are interacting with members of their own campus community. The platform is designed specifically for student needs: affordable second-hand goods, service exchanges, and local meetups that can occur safely on campus. By keeping the marketplace focused on SFSU, we can offer a convenient, campus-focused exchange medium to students.

Alli Trade is being developed by Team 17 at San Francisco State University as part of a collaborative software engineering project. Our team combines skills in software development, web design, and system deployment to build a reliable and scalable platform using modern web technologies. By building this platform ourselves, we gain real-world experience in designing, deploying, and maintaining a full-stack web application while creating a tool that can directly benefit the SFSU community. Our goal is to deliver a useful, student-focused marketplace that improves how students connect, trade, and support each other on campus.

2. Data Item Glossary

Term	Description
Unregistered User	A user that can browse the marketplace and view listings but must register and login to access other app features
Registered User	Someone accessing the marketplace that can also chat with other registered users , modify profile data, and create (and remove) listings
User	A term that encompasses both registered users and unregistered users .
Account	The registered users profile
Profile	The personal account the registered user makes when signing up for the marketplace for the first time. Contains the user's name, with optional visibility of the registered user's email, photo, major and graduation year or other active university role, and active listings . The registered users can add or delete posts from this page.
Group	Manages a set of permissions (i.e. users , posts , items) which provide CRUD access to database entities
User Group	See Group
Marketplace	Term used to refer to the webpage of active listings up for sale
Register	The act of creating an account to have full access to chat functions
Buyer	Registered User interested in purchasing item/service
Seller	Registered User listing item or service .
Product	A physical item put up for sale by a seller . It is a label of a product under the category
Service	A service or intangible item up for sale by a seller . It is a label of a product under the category
Item	A generalized term to easily refer to a product or service
Listing	Refers to active items in the marketplace
Price	The amount of money a seller is requesting for an item
Post	The act of completing an item's necessary parts to be visible listing and up for sale on the marketplace .

Listing Photo	Refers to the digital photo(s) of a listing .
Description	A short paragraph describing the item for sale, made by the seller when posting an item on the marketplace
Profile Data	Photo, major, graduation year
Admin	A registered user with heightened group privileges on the app. Admins can add/remove moderators and ban accounts , and have moderator privileges
Moderator	A registered user with heightened group privileges on the app. Moderators can flag accounts , suspend accounts , and hide listings
Super Admin	A registered user with heightened group privileges on the app. Super-Admins have admin and moderator privileges, and can enable or disable features of the app or shut down the app entirely.
Chat	The in-app function of messaging other registered users . Primarily used for buyer-seller communications
Message	The act of starting a chat with another registered user .
Category	A label applied to a listing by the registered user . This label is used to filter listings .

3. Functional Requirements

Priority 1

- 1.1. **Registered Users** are only allowed to use an email with @sfsu.edu to gain access to the app.
- 2.1. **Registered Users** shall create **profiles**, with their name, profile picture, major and graduation year or other university role
- 2.2. **Registered Users** shall post **listings** for **items (products or services)** they want to sell
- 2.3. **Registered Users** shall log in to their **accounts**
- 2.5. **Registered Users** shall log out of their **accounts**
- 5.1. **Registered Users** shall be able to create a **listing**
- 5.3. **Registered Users** shall be able to label a **listing** as a **product** or **service**, with a **price**, **picture(s)**, and a **description** for the **item** as well as label the **item** with 1 or more **categories**
- 3.2. **Unregistered Users** shall be able to **search** for **listings**
- 3.3. **Unregistered Users** shall be made to **register** before being able to message other **registered users**.
- 3.1. **Unregistered Users** shall be able to browse **listings** on the **marketplace** without creating an **account**.
- 6.1. **Users** shall be able to search for **products/services** using a search bar
- 6.3. **Users** shall be able to browse all **listings** with a set amount of **listings** per page
- 6.4. **Users** shall be able to change pages
- 6.5. **Users** shall be able to sort **listings** by **price**, **category**, **product**, or **service**
- 7.1. **Buyers** shall be able to **chat** with **sellers** via in-app text messaging

Priority 2

- 2.4. **Registered Users** shall update their **profile** information
- 4.2. **Admins** shall have **admin** access levels (**Moderator**, **Admin**, **Super Admins**)
- 4.3. **Moderators** shall be able to flag **accounts**, suspend **accounts**, hide **listings**
- 5.2. **Registered Users** shall be able to update (edit or remove) their current **listings**
- 6.2. **Users** shall be able to use **categories** to browse specific **products/services**

7.2. **Chats** shall be deleted when **listings** are deleted

Priority 3

4.4. **Moderators** shall be able to report accounts to **admins** for further actions

4.5. **Admins** shall be able to add/remove **moderators** and ban **accounts**

4.6. **Admins** shall have **moderator** privileges

4.7. **Super Admins** shall have **admin** and **moderator** privileges

4.8. **Super Admins** shall have the ability to remove **admins** and **moderators**

4. UI Storyboards

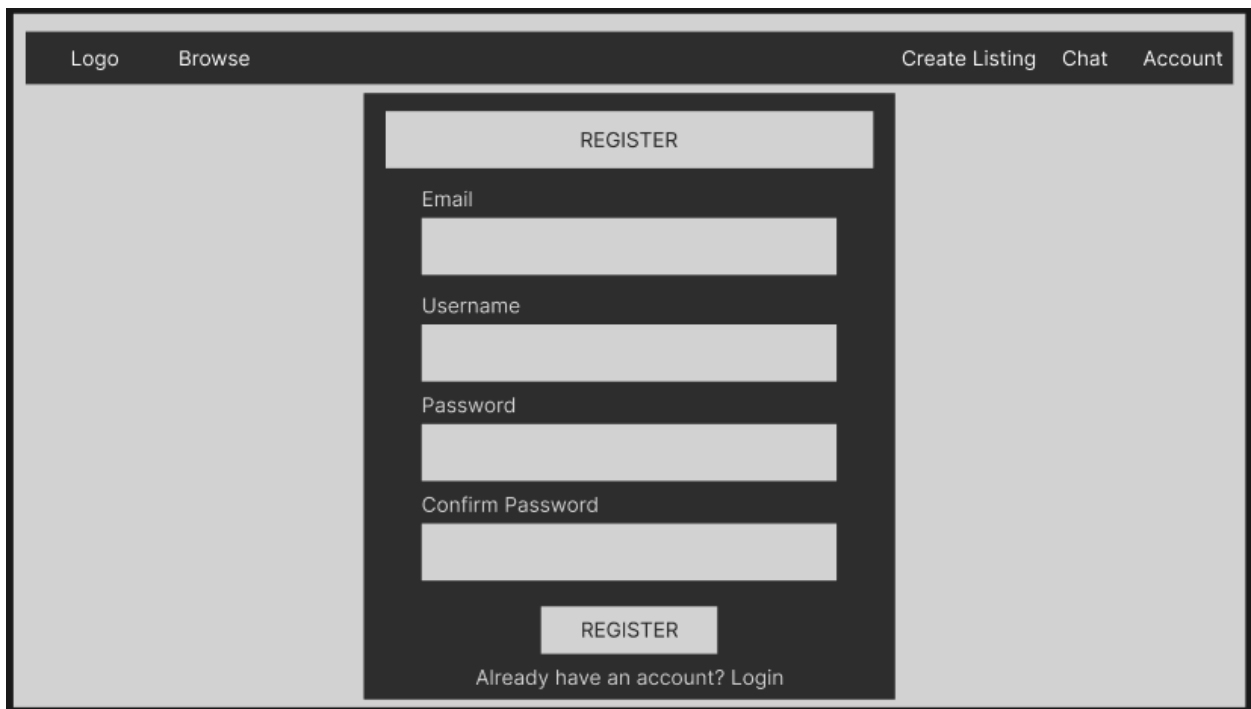
Use Case 1: User Creates a service listing

Marketplace Home page Mockup

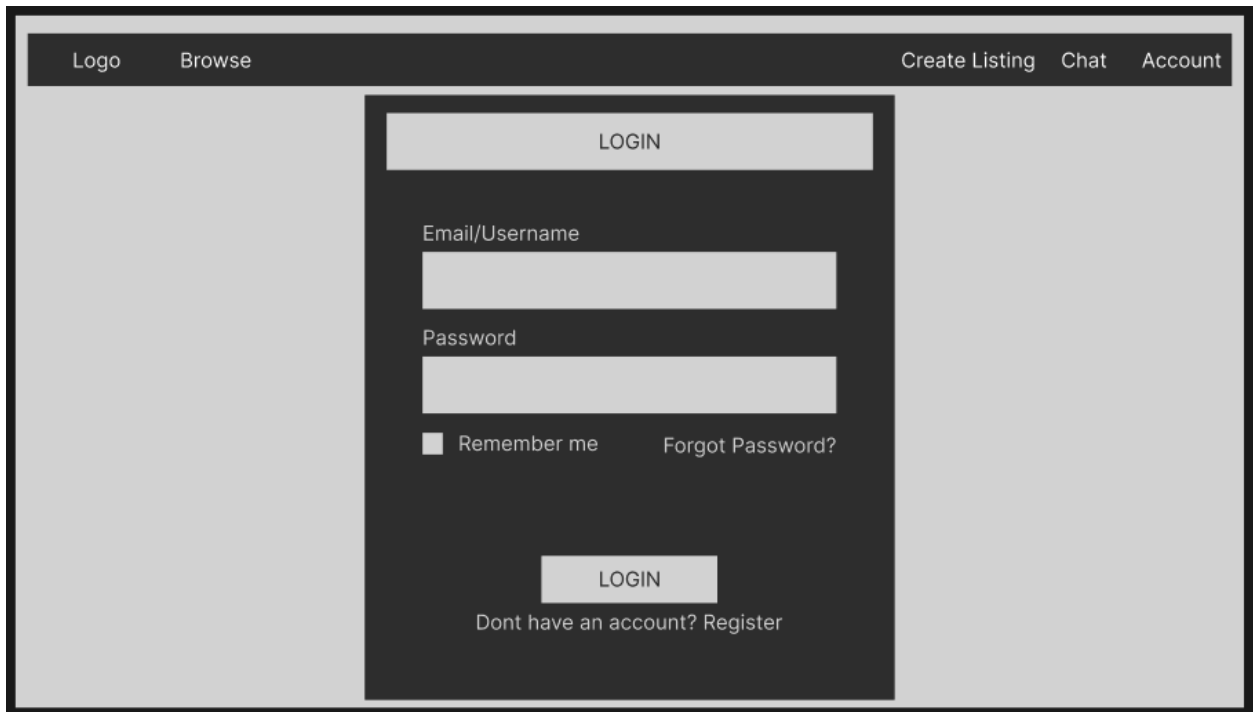
Context: User is browsing the marketplace home with only one listing



Registration Mockup



Login Mockup



A login form mockup with a dark gray background. At the top, a navigation bar contains 'Logo', 'Browse', 'Create Listing', 'Chat', and 'Account'. The login form is centered and features a 'LOGIN' button at the top. Below it are input fields for 'Email/Username' and 'Password'. A 'Remember me' checkbox and a 'Forgot Password?' link are positioned below the password field. At the bottom of the form is another 'LOGIN' button and a link that says 'Dont have an account? Register'.

Logo Browse Create Listing Chat Account

LOGIN

Email/Username

Password

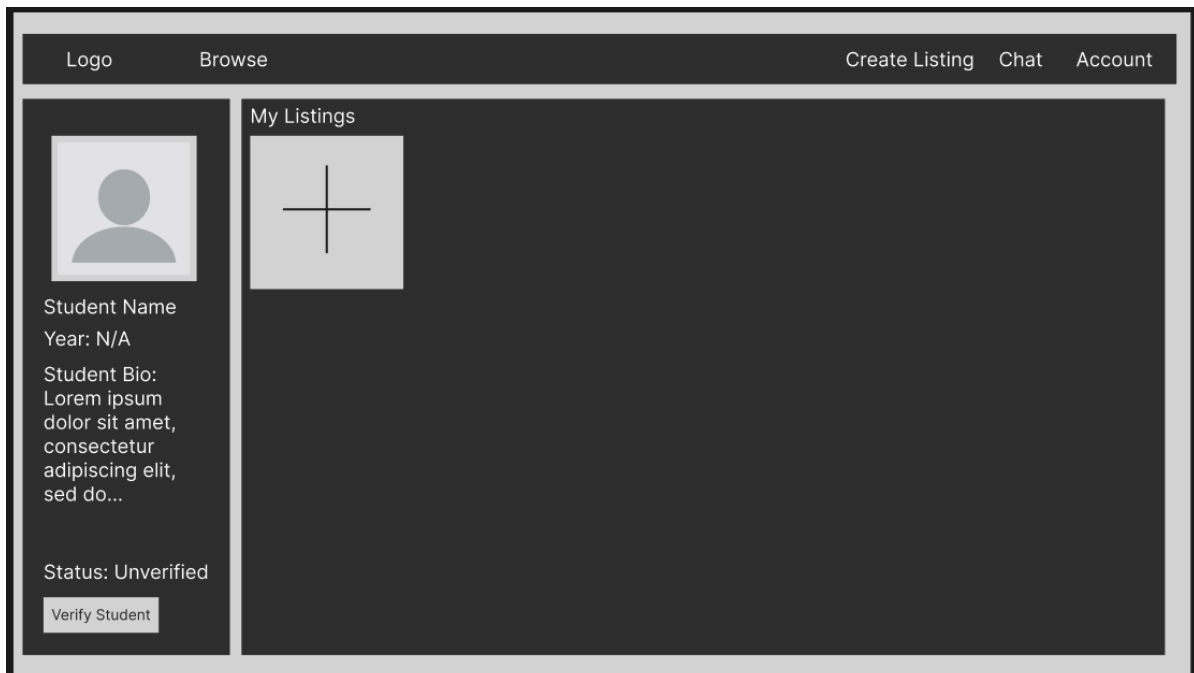
☐ Remember me Forgot Password?

LOGIN

Dont have an account? Register

Account Mockup

Context: Registered User is viewing their own profile page on an unverified student account



An account profile mockup with a dark gray background. The top navigation bar is identical to the login page. The profile section on the left includes a placeholder for a profile picture, the user's name 'Student Name', their year 'Year: N/A', a bio 'Student Bio: Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do...', and their status 'Status: Unverified' with a 'Verify Student' button. The main area is titled 'My Listings' and contains a large placeholder with a plus sign.

Logo Browse Create Listing Chat Account

My Listings

Student Name
Year: N/A
Student Bio:
Lorem ipsum
dolor sit amet,
consectetur
adipiscing elit,
sed do...

Status: Unverified
Verify Student

Create Service Mockup

Logo

Browse

Create Listing

Chat

Account

Create Service

Add Image

Publish Service

Title

Description

Price per hour

Category

Use case 2: user searches and browses listings

Marketplace Home page Mockup

Context: User is browsing the marketplace with only one listing



Search listing Mockup

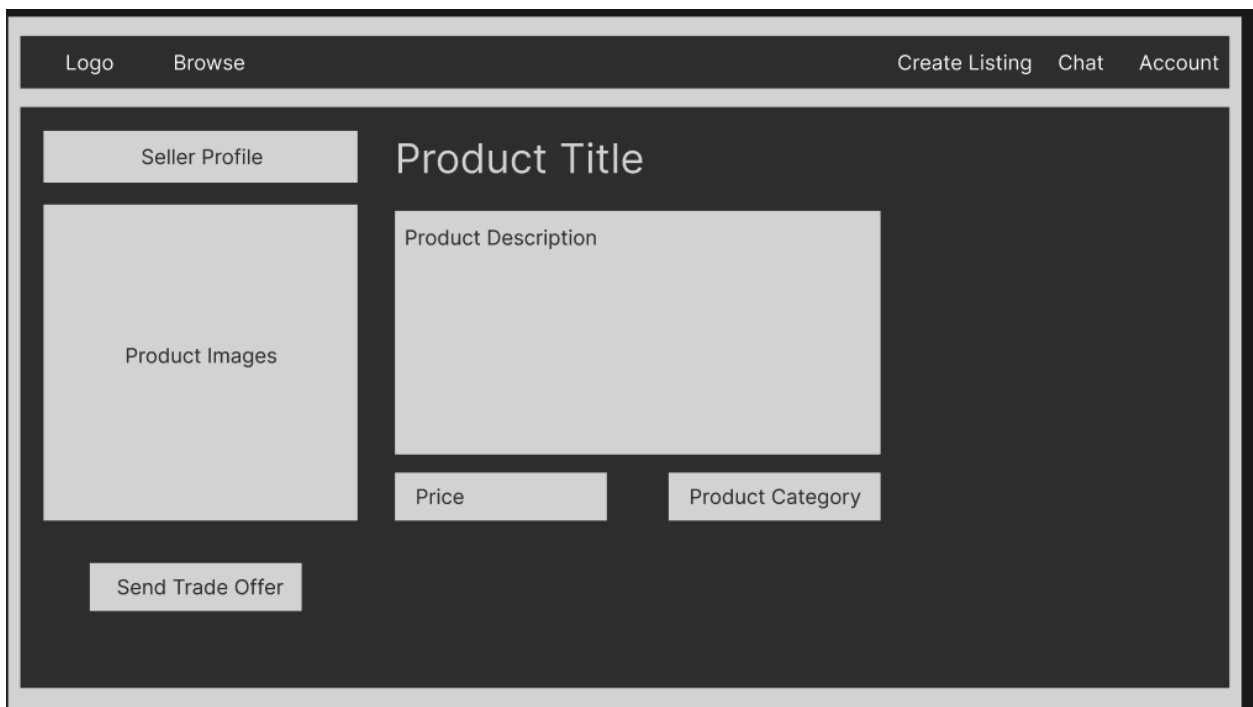
Context: User searches an item and has results display





View product/service Listing detail Mockup

Context: User has selected an item to view and is shows greater detail



Use Case 3: user creates and manages listing

Account Mockup

Context: User is viewing their own profile page on an unverified student account

The mockup shows a user profile page. At the top is a navigation bar with 'Logo', 'Browse', 'Create Listing', 'Chat', and 'Account'. The main content area is split into two columns. The left column contains a profile picture placeholder, the text 'Student Name' and 'Year: N/A', a bio 'Student Bio: Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do...', and a status 'Status: Unverified' with a 'Verify Student' button. The right column is titled 'My Listings' and contains a large square placeholder with a crosshair, indicating where a listing image would be.

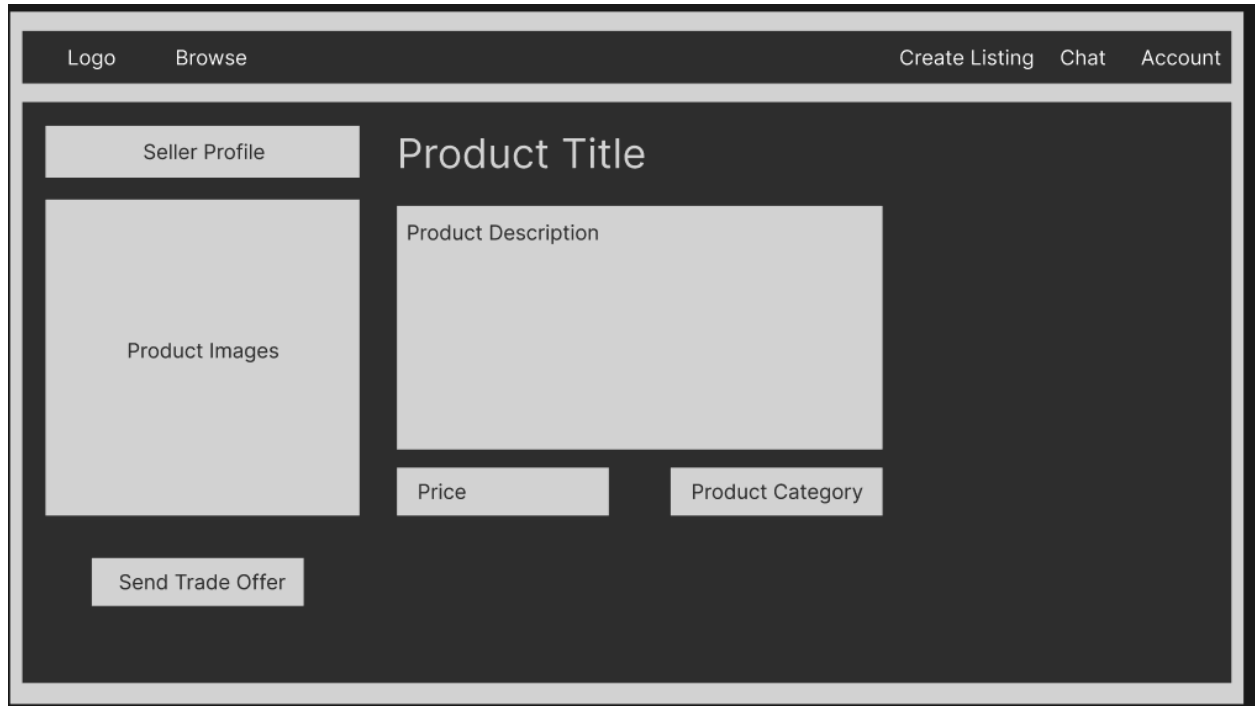
Create Listing Mockup

Context: User is creating a listing for a product/service they wish to sell

The mockup shows a 'Create Listing' form. It has the same top navigation bar as the account page. The form is divided into two main sections. The left section is titled 'Create Listing' and contains a large 'Add Image' placeholder and a 'Publish Listing' button. The right section contains input fields for 'Title', 'Description', 'Price', and 'Category'.

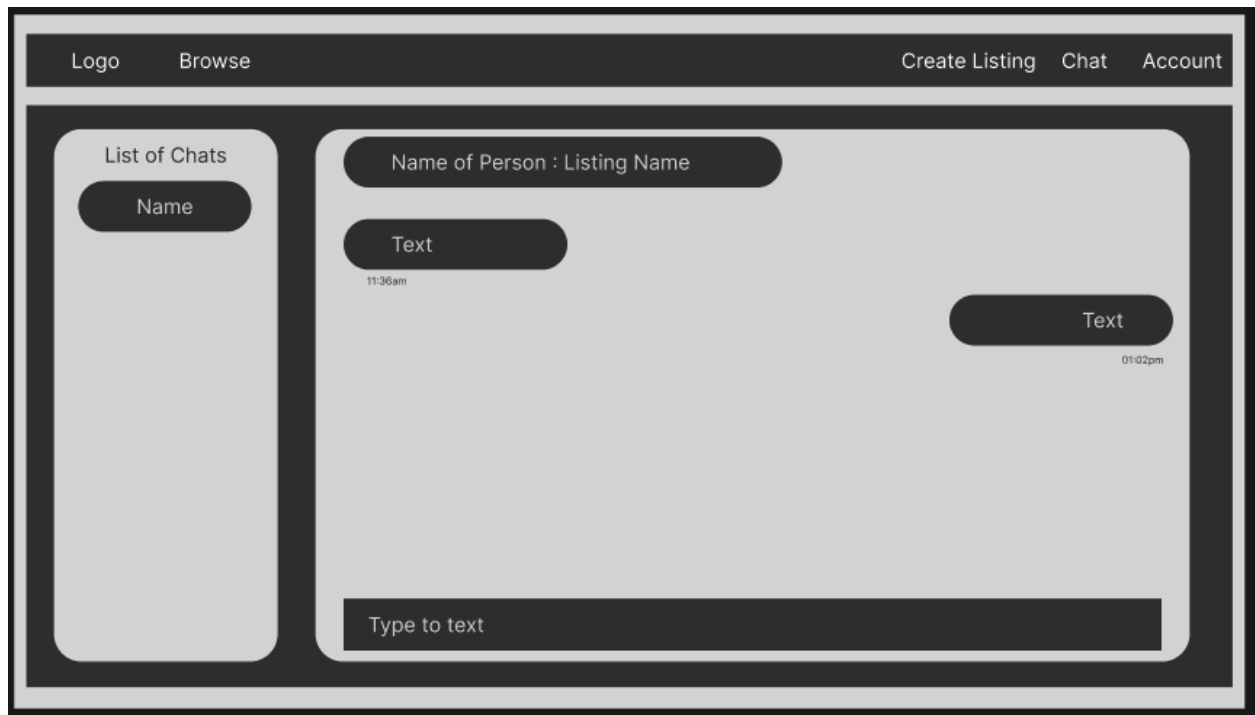
Use case 4: Buyer contacts seller

View product/service Listing detail Mockup



Chat Mockup

Context: User opens up the chat window for either checking for users who are interested in product, or users who are looking to sell said product



5. High level Architecture, Database Organization

DB organization

Database Table Title	Table Items	Descriptions
User	user_id (pk) sfsu_email first_name last_name role (fk) password_hash account_status created_at updated_at	Unique identifier for each user Verified SFSU email address User's first name User's last name Reference to roles Encrypted password Active or Suspended Account creation timestamp Tracks when listings or account is/are updated
Listing	listing_id (pk) title description price listing_type condition	Unique identifier for listing Listing title Detailed Description Listing Price Item or Service New, like-New, Good, Fair

	category_id(fk) seller id (fk) status created_at updated_at	Reference to category Reference to user who created listing Active, Sold, Removed Date of creation Last update timestamp of specific listing
Category	category_id(pk) category_name category_description	Unique category identifier Category name (Books, Electronics, Furniture) Option Description
Chat	message_id(pk) listing_id(fk) sender_id(fk) receiver_id(fk) message_text sent_at created_at deleted_at	Unique message identifier Associated listing Message sender Message recipient Content of the message Timestamp of message Timestamp of creation Timestamp of deletion
Role	role	User's role: admin, moderator, super_admin

Media storage

Images will be stored in the server file system, not as database BLOBs. The Image table stores only the relative file path (image_path) pointing to the file on disk. This approach is preferred because it avoids bloating the database and aligns with standard Django media file handling via MEDIA_ROOT. Standard image formats (JPEG, PNG) will be accepted. No video or audio is supported in this version currently, subject to change.

Search/filter architecture and implementation

Search Function will be implemented using SQL queries with the %LIKE operator for partial matching.

The user will be able to search for listings by **Title**, **Description**, and **Category**

Filtering will also be supported by category and price range.

Pagination will be implemented to limit the number of listings displayed per page.

All searchable fields will be indexed where appropriate to improve performance.

Algorithms/Processes

At this moment, no complex algorithms.

However, there will be some simple processes:

- Listing Prioritization
Listings are displayed in reverse chronological order (most recent first)
- Chat Handling
Messages are grouped by chat sessions between two users to maintain chat history

Framework/Tool Updates

No changes or updates to SW tools at this time, Subject to Change

6. Risks

Skills risks:

At this point, one potentially major risk regarding skills issues may come down to team knowledge. Some members associated with front-end and back-end, respectively, have admitted that they are not used to the frameworks we are currently using. This has yet to pose much of an issue at this time.

Skills Resolution:

We plan to approach this risk by having each team member spend time with the frameworks they are working on that they are also unfamiliar with to build knowledge and familiarity. It is up to individuals to educate themselves on the frameworks necessary to get the job done. Task assignment is adjusted to this within reason. However, we will all try our best to support one another with learning and jumping from team to team when needed.

Schedule risks:

The only known schedule risk at this time would be the lack of availability with coordinated FULL team meetings. Friday morning meetings have worked incredibly well for all members involved so far. But, limit our direct work on the application as a team to a few hours a week, leaving the in-between time for individual work on select portions of the app. This may pose an issue down the line.

Schedule resolutions:

Our current resolution for this is for the back-end and front-end members to meet as subgroups during their own respective availabilities. This allows team coordination and communication whilst having more flexibility with scheduling since the teams are smaller. This allows team members to stay in communication, on track with work, and then update the other half during our full team meetings.

Technical risks:

Many of the current technical challenges have not yet been approached. Some of the more complex features, such as bidding and messaging functionality, have not yet been broken down. As requirements become more technically sophisticated, there is increased risk of exceeding available development time.

There haven't been any additional deployments done that include interacting with the database, application, or content assets. The dependability of the deployment pipeline is unknown and might become an issue as we iterate more frequently in future milestones.

Technical resolutions:

Scoping of the technical requirements early on will help set more realistic expectations of what can be delivered by the due date. (This section probably be updated before M2 submission, because we will have a better sense of these features before it is due)

Teamwork risks:

One teamwork risk is the lack of experience working in large team based projects. Most of us are inexperienced when it comes to working in large and sub-teams, splitting up work, and actively doing check points. This can cause not only issues in making progress but also having ideas be implemented differently depending on the person. This can cause confusion and overlap which we already had experienced when creating our server.

Teamwork Resolutions:

We are utilizing a number of tools to improve communications, such as Discord and meeting schedulers. A regular weekly meeting assures that there is a forum in which active milestones can be discussed. Additional tools that integrate with GitHub, JIRA, and Discord will also improve issue tracking and keeping up with technical contributions.

Legal/content risks:

Mainly copyright from SFSU or users believing products/services are real offerings.

Legal/content Resolutions:

Risks will be avoided by including a disclaimer banner at the top of the page absolving individuals of responsibility; limited risk due to educational use-only of materials; use of video, images, and what would be considered production-grade user-generated content will be either free-to-use or synthetically generated using GPT tools.

7. Project Management

For project management, we use Jira to manage tasks and Discord for team communication. Milestones are assigned on Thursdays, and we will have a full team meeting on Friday mornings where we plan how to complete the upcoming milestone. During this meeting, we identify key tasks such as storyboarding and data glossaries. Tasks are then split into smaller, manageable components, such as Storyboard 1, 2, etc., and added to our Jira board.

Tasks are assigned collaboratively based on each member's role and strengths, with front-end and back-end developers handling their respective areas, and the team lead managing broader responsibilities such as the executive summary. We set a clear team deadline, which will be prior to the actual milestone deadline, and a check-in point to track progress. The team lead monitors task completion, while each member is responsible for updating their progress, completing assigned work, and seeking help when needed.

Although tasks are assigned individually, we maintain a collaborative team environment to support one another. This structured approach helps us stay organized, distribute work effectively, and ensure accountability throughout the project.

8. Sources/Gen AI

Version	Task / Used?	Rating	Explanation / Examples / Prompt / Comments
ChatGPT GPT-5.3 model	Functional Req. Priorities	Good	● Prompt: Based on our buy and sell app for sfsu students can you rank the following requirements in priority 1 to 3 ● Output was well formatted and had good reasoning behind priority type ● Satisfied with results no correction needed
...	...	...	● ... ● ...

9. Team Lead Checklist

So far all team members are fully engaged and attend team sessions when required

- *Done - **On Track** - Issue*

Team ready and able to use the chosen back and front-end frameworks and those who need to learn are working on learning and practicing

- ***Done** - On Track - Issue*

Team reviewed suggested resources before drafting Milestone 2

- ***Done** - On Track - Issue*

Team lead checked Milestone 2 document for quality, completeness, formatting and compliance with instructions before the submission

- ***Done** - On Track - Issue*

Team lead ensured that all team members read the final Milestone 2 document and agree/understand it before submission

- ***Done** - On Track - Issue*

Team shared and discussed experience with GenAI tools among themselves

- ***Done** - On Track - Issue*